

A Catalogue of Green Pea Galaxies in the Kilo-Degree Survey DR5 with the GP-YOLO11 Deep-Learning Framework

ABSTRACT

Green Pea galaxies (GPs) are compact and extreme star-forming systems that serve as important local analogs of high-redshift Ly α -emitting galaxies. Identifying these rare objects in wide-field optical surveys is challenging because their broadband colors can overlap significantly with those of stars and quasars. In this work, we develop GP-YOLO11, a customized deep-learning framework for the large-scale automated search for GPs in the Kilo-Degree Survey Data Release 5 (KiDS DR5). Rather than relying solely on photometric color cuts, the model analyzes four-channel $u/g/r/i$ image cutouts constructed from the KiDS multiband data and is optimized for the detection of compact astronomical sources. Comparative evaluations demonstrate that GP-YOLO11 outperforms baseline detection models, achieving a precision of 97.97%, a recall of 97.61%, and an mAP@0.5 of 98.36%. Applying the trained model to more than one million sources within the photometric redshift range of $0.1 < z < 0.4$, we obtain a catalog of 5,699 high-confidence GP candidates. By cross-matching these candidates with previously reported GP catalogs and with optical spectra from the Dark Energy Spectroscopic Instrument (DESI) and the Sloan Digital Sky Survey (SDSS), we recover 59 known GPs and identify 61 newly spectroscopically confirmed GPs. These objects lie in the extreme star-forming region of the Baldwin-Phillips-Terlevich (BPT) diagram and show no evidence of active galactic nucleus (AGN) contamination. Our results demonstrate that deep-learning-based image analysis can effectively alleviate the limitations of traditional color-selection methods and provide an efficient approach for expanding samples of rare extreme emission-line galaxies (EELGs) in large imaging surveys.

Keywords: Galaxies (573) — Emission line galaxies (459) — Galaxy evolution (594) — Astronomy data analysis (1858)

1. INTRODUCTION

Green Pea galaxies (GPs) are a rare class of compact, extreme emission-line galaxies (EELGs) experiencing intense star formation. They were first identified by citizen scientists in the Galaxy Zoo project because of their unresolved appearance and characteristic green color in Sloan Digital Sky Survey (SDSS) composite images (C. Cardamone et al. 2009). GPs exhibit distinctive physical properties, including high specific star formation rates (sSFR), relatively low stellar masses ($\sim 10^8 - 10^{10} M_{\odot}$), and subsolar metallicities (R. Amorín et al. 2010; Y. I. Izotov et al. 2011). Owing to these properties, they are widely regarded as important nearby analogs of high-redshift Ly α emitters and Lyman-break galaxies (A. Henry et al. 2015; H. Yang et al. 2017). In addition, some GPs are known to leak ionizing Lyman continuum photons, making them valuable local laboratories for investigating the physical conditions associated with cosmic reionization and the evolution of early galaxies (A. E. Jaskot & M. S. Oey 2013; Y. I. Izotov et al. 2016; Y. I. Izotov et al. 2016).

The identification of GPs in modern wide-field surveys remains challenging. Their rarity and compact morphology make them difficult to isolate efficiently in massive imaging datasets, while traditional visual inspection is no longer practical at current survey scales (D. Baron 2019). Previous searches have often relied on photometric color-selection criteria designed to capture the strong [O III] emission that characterizes these systems (C. Cardamone et al. 2009; Y. I. Izotov et al. 2011). Although effective in some cases, such methods can suffer from significant degeneracies in broadband color space, where stars and quasars may overlap with genuine GPs. As a result, rare-object searches in large survey datasets increasingly benefit from machine-learning methods that can move beyond simple parameter cuts and exploit higher-dimensional information.

Recent studies have shown that deep-learning approaches can be effective for identifying GPs and related rare emission-line systems in large photometric datasets (P. Yang et al. 2026). In particular, P. Yang et al. (2026) applied the MobileMamba-ECA architecture to perform a large-scale search for GPs in the Dark Energy Spectroscopic Instrument (DESI) Legacy Imaging Surveys (A. Dey et al. 2019). This work demonstrated the promise of deep learning for mitigating color degeneracies in survey data with limited band coverage. However, methods optimized for one survey are not necessarily optimal for another, especially when the available image information differs substantially in both bandpass coverage and spatial resolution.

This consideration is particularly relevant for the Kilo-Degree Survey Data Release 5 (KiDS DR5). Compared with the DESI Legacy Imaging Surveys, KiDS provides four-band optical imaging (u, g, r, i) together with higher spatial resolution, with a median r -band seeing of about 0.7 arcsec (A. H. Wright et al. 2024). These characteristics offer an opportunity to exploit not only color information but also fine-scale morphological structure in compact galaxies. These differences motivate a survey-specific framework that can exploit both the multiband photometric information and the spatial structure available in KiDS images. To take full advantage of the KiDS data quality, a dedicated framework is needed for the identification of compact GP candidates from multiband imaging (A. H. Wright et al. 2024).

In this work, we present GP-YOLO11, a customized deep-learning framework developed for the large-scale automated search for GPs in KiDS DR5 (G. Jocher & J. Qiu 2024). We construct a training sample from previously reported GPs, apply the trained model to KiDS sources within the redshift range relevant for GP searches, and distinguish throughout this work between spectroscopically confirmed GPs and photometrically selected GP candidates. The resulting catalog is intended to provide both a validated subsample of newly confirmed GPs and a larger list of high-confidence GP candidates for future spectroscopic follow-up.

This paper is organized as follows. Section 2 describes the KiDS DR5 data, sample construction, and data augmentation procedures. Section 3 presents the GP-YOLO11 framework and its methodological details. Section 4 evaluates the model performance and provides ablation analyses. Section 5 reports the spectroscopic validation results and the resulting GP candidate catalog. Section 6 summarizes our conclusions.

2. DATA

2.1. The Kilo-Degree Survey Data Release 5

KiDS (J. T. A. de Jong et al. 2013) is a public optical imaging survey conducted with the VLT Survey Telescope (VST; M. Capaccioli & P. Schipani 2011) and the OmegaCAM camera at the European Southern Observatory. In this work, we use the fifth public data release of KiDS (KiDS DR5) (A. H. Wright et al. 2024), which includes all survey tiles and covers a total area of 1347 deg^2 . KiDS provides imaging in four broad optical bands, namely u , g , r , and i , with relatively uniform depth and high image quality. In particular, the median seeing in the r band is approximately 0.7 arcsec, and the 5σ limiting magnitude reaches 24.8. These characteristics make KiDS DR5 a suitable imaging dataset for searches for faint and compact extragalactic sources, including GP candidates.

2.2. Data Set Construction and Sample Selection

The goal of this study is to identify GPs in the KiDS DR5 survey using a deep-learning-based object detection framework. To train and evaluate the model, we constructed a labeled data set consisting of two subsets: a positive sample of spectroscopically confirmed GPs and a negative sample of non-GP sources.

For the positive sample, we collected previously reported spectroscopically confirmed GPs from the literature and cross-matched them with public spectroscopic resources, including SDSS (D. G. York et al. 2000) and Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST; X.-Q. Cui et al. 2012), to assemble a unified GP source list (e.g., T. Jiang et al. 2019; S. Liu et al. 2022; P. Yang et al. 2026). Duplicate entries were removed before cross-matching the remaining sources with the KiDS DR5 footprint. We then applied a photometric-redshift criterion of $0.1 \leq z_B \leq 0.4$, where z_B is the photometric redshift estimate provided by the KiDS catalog. Following this selection, we extracted image cutouts centered on the right ascension and declination of each target. The extracted multiband cutouts have a native pixel scale of $0.21'' \text{ pixel}^{-1}$ with dimensions of 512×512 pixels in each band. We used the aligned u , g , r , and i cutouts to construct four-channel image tensors, which were used as the input to the detection model. This cutout size was adopted to retain sufficient local context around each compact source while preserving the native KiDS image quality. Consequently, we obtained an initial data set of 305 positive GP images. Given that

this sample size was insufficient for training the deep-learning model, we further expanded the positive sample through data augmentation, as described in Section 2.3.

For the negative sample, we constructed a sample to account for potential morphological confusion in the identification of GPs. The sample consists of three categories of potential contaminants: 1,000 stars, 1,000 quasars (QSOs), and 1,000 spectroscopically classified non-GP galaxies. These objects were selected because they can appear compact or unresolved in ground-based imaging, making them potential contaminants for GP searches. The sources were drawn from spectroscopic catalogs to ensure reliable class labels. While the labels were taken from external catalogs, the corresponding KiDS DR5 image cutouts were used for model training, ensuring that the negative samples share the same survey depth, pixel scale, and image quality as the positive GP samples. Using this retrieval protocol, we obtained a total of 3,000 negative images.

The full labeled data set was randomly divided into training, validation, and test subsets with a ratio of 8:1:1. The split was performed at the source level before augmentation, such that augmented versions of the same original image appeared only in the training set. To evaluate the model on original observational data, the validation and test sets were restricted to unaugmented images. Each of the validation and test sets contains 330 images, including 30 GPs and 300 negative samples. The data augmentation described in the following section was applied exclusively to the training split to address class imbalance and reduce overfitting.

For the subsequent large-scale search, we constructed a photometrically pre-selected parent sample from the KiDS DR5 catalog. This selection was based on general criteria involving photometric redshift, signal-to-noise ratio, and compactness, in order to retain sources in the redshift regime where GPs are expected to be efficiently identified. The trained GP-YOLO11 model was then applied to this parent sample for candidate identification, as described in Section 5.

2.3. Data Augmentation

Following standard practice in astronomical image augmentation (N. Sedaghat & A. Mahabal 2018), we applied data augmentation only to the training split because the positive training sample was limited to 245 confirmed GPs. This strategy was adopted to increase sample diversity and reduce the risk of overfitting. After augmentation, the positive training sample was expanded to 2,400 images, resulting in an approximately 1:1 balance between positive and negative samples in the training set. The augmentation procedures used in this work include geometric transformations, mosaic augmentation, and noise injection, as described below and illustrated in Figure 1. For the four-channel inputs, all geometric transformations were applied identically to the u , g , r , and i channels to preserve the spatial alignment among the bands.

- **Geometric Transformation:** To account for variations in source orientation and position, the training images were subjected to random rotations covering the full range of $0^\circ - 360^\circ$, as well as random horizontal and vertical flips.
- **Mosaic Augmentation:** We applied mosaic augmentation during training following the YOLO framework, in which multiple cutout images were combined to generate more complex training scenes. This procedure is intended to improve the model’s ability to detect compact targets in diverse image backgrounds.
- **Noise Injection:** Although the images were drawn from the KiDS DR5 survey, variations in observational depth, sky background, and exposure coverage remain across survey tiles and detector positions. To simulate such variations, we introduced random synthetic Poisson and Gaussian noise. This augmentation improves the robustness of the model to variations in signal-to-noise ratio.

3. METHOD

This section describes the architecture of the proposed detection framework for the automated identification of GPs in wide-field survey data. We first introduce the YOLOv11 architecture, followed by a detailed explanation of the Receptive-Field Attention Convolution (RFACnv). We then describe how these components are integrated into the optimized model, together with the loss functions used for training.

3.1. YOLOv11 Framework

We adopt the YOLOv11 (You Only Look Once, version 11) architecture as the base detection framework (J. Redmon et al. 2016; G. Jocher & J. Qiu 2024). YOLOv11 is a one-stage object detection framework that processes an entire

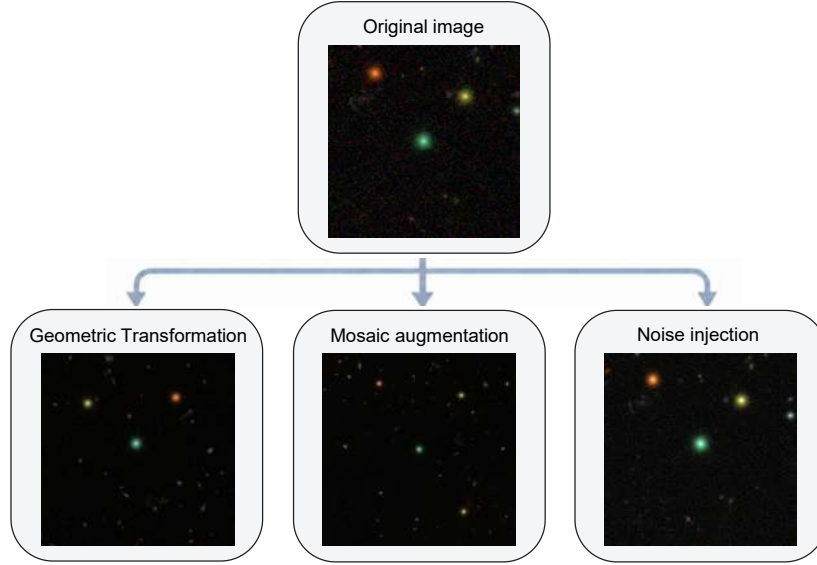


Figure 1. Examples of the data augmentation strategies used in this work. The top panel shows the original multiband cutout of a representative GP from the KiDS DR5 survey. The three bottom panels illustrate example outputs after applying geometric transformation, mosaic augmentation, and noise injection, respectively.

image in a single forward pass, jointly predicting bounding boxes and class probabilities. Specifically, we utilize the YOLOv11n (Nano) variant, which has substantially fewer parameters and lower computational cost than the larger variants in the same model family.

The backbone of YOLOv11n is a modified CSP (Cross Stage Partial) Darknet structure (C.-Y. Wang et al. 2020), which efficiently extracts hierarchical morphological features from input images. We chose the lightweight YOLOv11n variant because GPs are rare and typically occupy only a small fraction of a 512×512 pixel image. Under these conditions, a compact architecture helps reduce the risk of overfitting while remaining suitable for compact-target detection with a limited training set.

3.2. Receptive-Field Attention Convolution (RFACnv)

Detecting GPs is challenging due to their visual similarity to point-like contaminants, such as stars and quasars (QSOs). In conventional convolutional layers, the same kernel weights are applied across all spatial locations (X. Zhang et al. 2023). These standard layers may be less effective at capturing subtle spatial gradients and centrally concentrated light distributions in compact sources. These structural differences provide important cues for distinguishing a compact galaxy from a true point source.

To address this limitation, we introduce Receptive-Field Attention Convolution (RFACnv), which incorporates a spatial attention mechanism into the convolution operation (X. Zhang et al. 2023). Given an input tensor $X \in \mathbb{R}^{C \times H \times W}$, the attention-weighted feature map F_{attn} is computed as:

$$F_{\text{attn}} = \text{Softmax}(\text{GroupConv}(\text{AvgPool}(X))) \otimes \text{Unfold}(X), \quad (1)$$

where $\text{Unfold}(\cdot)$ extracts local patches by sliding a window over the input feature map, and $\text{GroupConv}(\cdot)$ generates position-dependent attention weights within the receptive field.

The weighted receptive-field features are then reconstructed as

$$X_{\text{re}} = \sum_{i=1}^{k^2} a_i \cdot x_i. \quad (2)$$

where k denotes the kernel size, x_i represents the i -th local feature within the receptive field, and a_i is the corresponding attention weight. Through this reconstruction process, RFACnv enhances the contribution of informative local structures while suppressing less relevant background features.

We adopt RFACnv because it relaxes the rigid spatially invariant behavior of conventional convolutions and enables more flexible local feature modeling. Since GPs typically exhibit high central surface brightness and rapidly declining outer light profiles, this mechanism is well suited to capturing the compact morphological patterns that distinguish them from stellar contaminants.

3.3. Proposed Model: GP-YOLO11

As illustrated in Figure 2, our proposed model, GP-YOLO11, integrates the RFACnv modules and an NWD-based detection head into the lightweight YOLOv11n baseline. We embed the RFACnv blocks within the neck of the network. This design enhances the fusion of fine-grained spatial features within the Path Aggregation Network (PAN).

Although the KiDS cutouts used in this work are centered on catalogued sources, we formulate the task as a detection-based candidate-identification problem rather than as a blind object-detection search over full survey tiles. This formulation is adopted because each cutout may contain neighboring sources, blended structures, background fluctuations, or residual artifacts. The YOLO framework allows the model to localize the compact GP-like emission region within each cutout and assign a confidence score to the detected source, rather than relying only on a global image-level label.

We optimize the detection head using the Normalized Wasserstein Distance (NWD) metric to address the specific challenges of tiny object localization (J. Wang et al. 2023). Standard Intersection over Union (IoU) metrics are highly sensitive to minor pixel deviations for extremely small targets. A single-pixel shift in a 5×5 pixel bounding box can cause a substantial drop in the IoU value. This effect can make the regression process unstable during training.

NWD alleviates this problem by modeling both the predicted and ground-truth bounding boxes as two-dimensional Gaussian distributions. A standard bounding box $B = (c_x, c_y, w, h)$ can be represented as a Gaussian distribution $\mathcal{N}(\mu, \Sigma)$. The model defines the mean vector μ as the center coordinates $[c_x, c_y]^T$. The model defines the covariance matrix Σ as the spatial extent $\text{diag}(w^2/4, h^2/4)$.

The simplified second-order Wasserstein distance W_2^2 between the predicted box B_p and the ground-truth box B_g is expressed as:

$$W_2^2(B_p, B_g) = (c_{x,p} - c_{x,g})^2 + (c_{y,p} - c_{y,g})^2 + \frac{(w_p - w_g)^2 + (h_p - h_g)^2}{4}, \quad (3)$$

This distance is then normalized into a scale-invariant exponential form:

$$NWD(B_p, B_g) = \exp\left(-\frac{\sqrt{W_2^2(B_p, B_g)}}{C}\right), \quad (4)$$

The constant C serves as a dataset-specific scaling factor. The corresponding regression loss is defined as $1 - NWD$.

We adopt NWD because it is well suited to the observational properties of astronomical sources. Genuine GPs generally do not exhibit sharp physical boundaries in survey images, and their observed light distributions are further smoothed by atmospheric seeing and instrumental effects, making Gaussian-based localization more appropriate for compact sources without sharply defined boundaries. NWD provides smooth and continuous gradients for these tiny objects even when the proposed bounding boxes do not overlap physically.

The complete network takes $512 \times 512 \times 4$ four-channel image tensors constructed from the KiDS u , g , r , and i bands as input. To accommodate the four-band KiDS input, the first convolutional layer of the YOLOv11n backbone was modified from three input channels to four input channels, while the remaining network architecture was kept unchanged. We train GP-YOLO11 using a refined composite loss function:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{NWD}} + \lambda_2 \mathcal{L}_{\text{cls}} + \lambda_3 \mathcal{L}_{\text{dfl}}. \quad (5)$$

The regression term ($\mathcal{L}_{\text{NWD}}$) is used to optimize spatial localization of compact GP-like sources. The classification loss ($\mathcal{L}_{\text{cls}}$) employs Binary Cross-Entropy to distinguish GP candidates from non-GP sources and background artifacts. We apply a confidence threshold ($\text{conf} > 0.8$) during the final inference stage to construct the high-confidence image-selected candidate catalog. This threshold is used for candidate prioritization in the source-centered KiDS search, while spectroscopic validation is required to confirm the physical nature of the selected candidates.

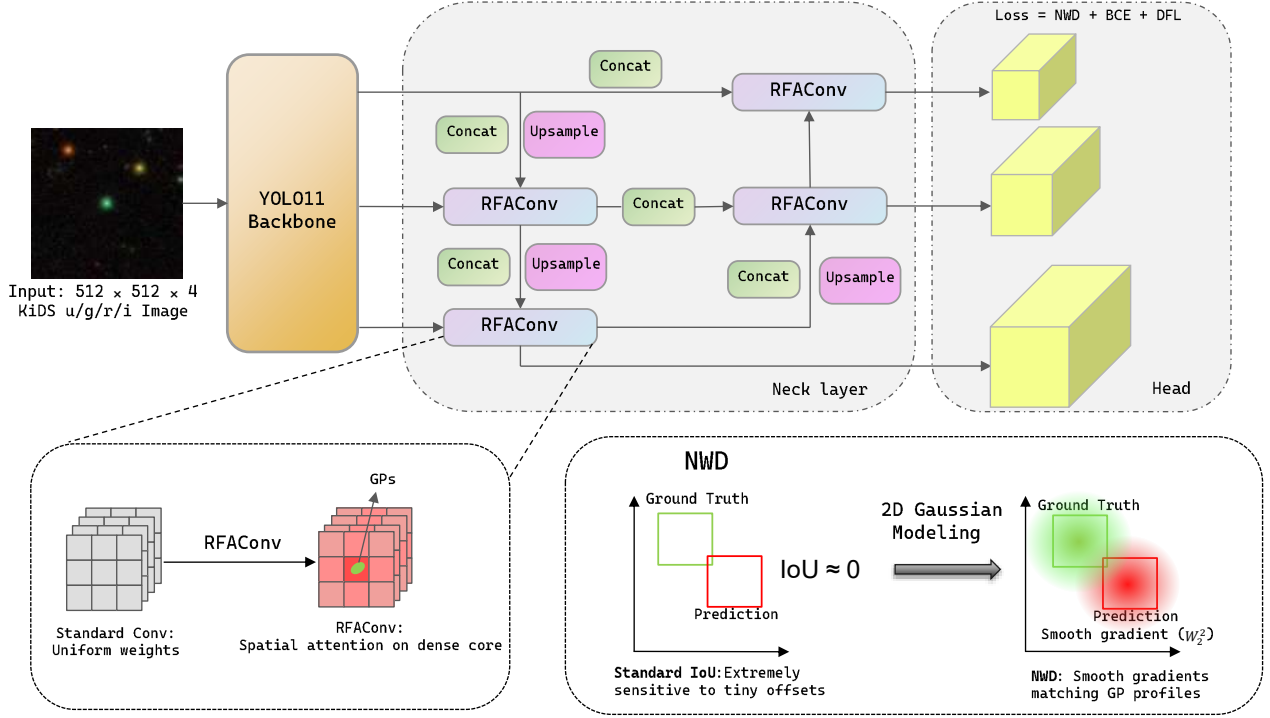


Figure 2. GP-YOLO11 model structure. The model consists of three parts: a YOLO11n backbone for extracting hierarchical morphological features from $512 \times 512 \times 4$ multi-band images, neck layers integrating Receptive-Field Attention Convolution (RFACConv) modules for adaptively focusing on the dense cores of GP candidates, and a detection head employing the Normalized Wasserstein Distance (NWD) loss for modeling tiny astronomical sources as two-dimensional Gaussian distributions.

4. EXPERIMENTAL RESULTS

This section presents the quantitative performance evaluation of the GP-YOLO11 framework. We first define the evaluation metrics used for source-level GP candidate identification. We then compare GP-YOLO11 with several complementary baselines, including a traditional photometric color-cut selection method, a catalog-based machine-learning classifier, an image-level CNN classifier, and representative object-detection architectures. Finally, we conduct an ablation study to isolate the impact of the proposed architectural modifications.

4.1. Evaluation Metrics

Because the comparison includes traditional photometric selection, catalog-based machine-learning classification, image-level classification, and object-detection models, we evaluate all methods as source-level binary GP candidate identification approaches. For the object-detection models, a source is classified as a GP candidate if the model produces at least one GP-like detection above the adopted confidence threshold in the corresponding cutout; otherwise, it is classified as a non-GP source.

We use Precision, Recall, and F1-score as the main comparison metrics:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (7)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (8)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. Precision measures the purity of the selected GP candidates, while Recall measures the fraction of true GPs recovered by each method. The F1-score provides a balanced measure of Precision and Recall, which is particularly useful for the imbalanced test set used in this work.

For detection-specific evaluation and ablation analysis, we also report the mean Average Precision at an Intersection over Union threshold of 0.5 ($mAP@0.5$). The Average Precision (AP) and mAP are defined as

$$AP = \int_0^1 P(R) dR, \quad (9)$$

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i, \quad (10)$$

where $N = 1$ for our single-class GP detection task.

We utilize a fixed IoU threshold of 0.5 to compute $mAP@0.5$ during testing. Although the training stage uses the NWD metric for bounding-box regression of tiny targets, evaluation is still performed using the standard IoU-based criterion for comparison with baseline models. The reported $mAP@0.5$ therefore serves as the main detection metric in our experiments.

4.2. Model Performance

The GP-YOLO11 framework achieves the best overall performance on the independent test set, reaching a Precision of 97.97%, a Recall of 97.61%, and an F1-score of 97.79%. These results indicate that the proposed method can effectively identify compact GP candidates from source-centered KiDS multiband cutouts.

To provide a more complete evaluation, we compare GP-YOLO11 with several types of baselines relevant to GP searches. First, we include a traditional color-cut selection baseline based on the KiDS $(g-r)$ vs. $(r-i)$ color-color plane. Following previous color-selection studies of Green Pea galaxies (C. Cardamone et al. 2009), we adopt the empirical criterion

$$(g-r) \geq (r-i) + 0.5 \quad (11)$$

as the photometric-selection baseline. Sources satisfying this criterion are classified as GP candidates, while the remaining sources are classified as non-GPs. This baseline represents the conventional broadband-color approach for identifying GP-like systems. Second, we include XGBoost as a catalog-based machine-learning baseline using KiDS photometric information. Third, we train a simple CNN classifier on the same source-centered KiDS cutouts to evaluate whether a pure image-level classification formulation can achieve comparable performance. Finally, we compare GP-YOLO11 with representative object-detection architectures, including Faster R-CNN (S. Ren et al. 2015), SSD (W. Liu et al. 2016), EfficientDet (M. Tan et al. 2020), and YOLOv8n (G. Jocher et al. 2023). For a fair comparison with the photometric-selection and classification baselines, all detection models are evaluated as source-level binary classifiers: a source is classified as a GP candidate if the model produces at least one GP-like detection above the adopted confidence threshold, and as a non-GP otherwise. Table 1 summarizes the results.

The color-cut selection baseline achieves a Recall of 100.00%, indicating that the adopted broadband-color criterion can recover the known GPs in the test set. However, its Precision is only 42.25%, resulting in a relatively low F1-score of 59.41%. This demonstrates the severe contamination problem caused by the overlap between GPs and non-GP sources in the KiDS $(g-r)$ vs. $(r-i)$ color space. The XGBoost baseline improves the Precision to 82.14%, showing that catalog-level photometric information provides more discriminating power than a simple color cut. Nevertheless, its performance remains below that of image-based methods, suggesting that catalog photometry alone does not fully capture the compact morphology and local image structure of GP candidates.

The simple CNN classifier reaches an F1-score of 93.55%, confirming that multiband KiDS image cutouts contain useful information for distinguishing GPs from contaminants. However, GP-YOLO11 still outperforms this pure image-classification baseline. This improvement suggests that the detection-based formulation, which explicitly localizes compact GP-like emission regions within each cutout, provides additional discriminating power beyond global image-level classification. Among the object-detection baselines, YOLOv8n gives the strongest performance, with an F1-score of 94.72%, but it remains below GP-YOLO11. Overall, GP-YOLO11 achieves the highest Precision, Recall, and F1-score among all tested methods, demonstrating the effectiveness of combining multiband image information, localized compact-source detection, RFACnv-based feature enhancement, and NWD-based localization.

4.3. Ablation Study

An ablation study was conducted to isolate and quantify the individual contributions of our architectural modifications. We constructed four model configurations based on the baseline YOLOv11n. We trained all variants using the

Table 1. Performance Comparison between Traditional Photometric Selection, Catalog-based Machine Learning, Image-classification Baseline, Object-detection Baselines, and GP-YOLO11

Method	Precision (%)	Recall (%)	F1-score (%)
Color-cut Selection	42.25	100.00	59.41
XGBoost	82.14	92.00	86.79
Simple CNN classifier	90.63	96.67	93.55
Faster R-CNN	92.13	91.73	91.93
SSD	92.56	92.31	92.43
EfficientDet	94.82	93.97	94.39
YOLOv8n	94.64	94.80	94.72
GP-YOLO11	97.97	97.61	97.79

same training, validation, and test splits, together with identical hyperparameter settings and random seeds, to ensure a consistent comparison. Table 2 summarizes the quantitative performance of each configuration.

The baseline YOLOv11n model achieves an $mAP@0.5$ of 95.78%. The integration of the RFACnv module (Variant 1) increases the Recall from 95.47% to 96.45%. This result suggests that the spatial attention mechanism helps the model recover more true GP sources, thereby improving recall.

Replacing the standard IoU metric with the NWD loss (Variant 2) increases the Precision and $mAP@0.5$ scores to 96.62% and 97.48%, respectively. By adopting a Gaussian-based formulation for tiny-target localization, NWD provides smoother optimization behavior for bounding-box regression of tiny astronomical sources.

Our proposed GP-YOLO11 framework combines both the RFACnv module and the NWD-based regression strategy. This complete configuration achieves the best performance among all tested variants, reaching an $mAP@0.5$ of 98.36%.

Table 2. Ablation Study of the Proposed Modules on the GP Detection Dataset. Checkmarks (✓) indicate that the corresponding module is included.

Model	RFACnv	NWD Loss	Precision (%)	Recall (%)	mAP@0.5 (%)
Baseline (YOLOv11n)			96.19	95.47	95.78
Variant 1	✓		96.89	96.45	96.45
Variant 2		✓	96.62	96.71	97.48
GP-YOLO11	✓	✓	97.97	97.61	98.36

5. NEW GP DISCOVERIES IN KIDS

5.1. Parent Sample and Candidate Identification

We performed a large-scale search for GP candidates within a photometrically pre-selected KiDS DR5 parent sample. We first applied photometric and quality constraints to the KiDS DR5 catalog to reduce the initial search space while retaining sources in the redshift regime relevant for GP identification. Specifically, we selected sources within the photometric-redshift range of $0.1 < z_B < 0.4$, based on the photometric redshift estimates provided by the KiDS DR5 catalog. This selection yielded a parent sample of 1,048,576 optical image cutouts for subsequent model inference.

The trained GP-YOLO11 model was applied to the full parent sample. Using a confidence threshold of > 0.8 , this process yielded 5,699 high-confidence image-selected candidates with GP-like properties, forming a compact catalog for subsequent spectroscopic follow-up. To evaluate these candidates, we cross-matched their coordinates with existing GP reference catalogs and the SIMBAD astronomical database. This successfully recovered 59 previously reported GPs, demonstrating the model’s capability to identify known compact emission-line galaxies within KiDS imaging. We further searched for archival spectroscopy among the 5,699 candidates using DESI DR1 and SDSS. Visual inspection of the available one-dimensional spectra confirmed 61 objects as newly identified GPs that had not been previously reported in the literature. The basic catalog information for these 61 newly identified GPs is provided in Appendix A.

In summary, the final candidate catalog consists of 59 recovered known GPs and 61 newly spectroscopically confirmed GPs. The remaining candidates are retained as promising targets awaiting future spectroscopic observations.

The whole candidate catalog reserves the original KiDS DR5 database columns, including source identifiers, astrometric information, multiband photometric measurements, morphological parameters, quality flags, photometric-redshift estimates, and derived physical quantities. In addition, several columns defined in this work are included, such as the candidate coordinates, the GP-YOLO11 confidence score, and the validation label. This format keeps the catalog directly traceable to the KiDS DR5 database while providing the information needed for candidate ranking and future spectroscopic follow-up. It is available from <https://github.com/CXYhaha/GP-YOLO11>.

5.2. Spectroscopic Confirmation of Newly Identified GPs

In this work, we use the term GP galaxy to refer to a compact, low-redshift, [O III]-dominated extreme emission-line galaxy. Following the original observational picture of GPs as compact systems with unusually strong [O III] emission, our spectroscopic confirmation is based on three main criteria. First, the source should lie within the redshift range targeted by our KiDS search. Second, it should exhibit compact morphology in the KiDS imaging and show strong nebular emission dominated by the [O III] $\lambda 5007$ line, with rest-frame $\text{EW}([\text{O III}] \lambda 5007) > 100 \text{ \AA}$. Third, it should display prominent Balmer emission lines and, when the required diagnostic lines are available, be classified as a star-forming system in the Baldwin–Phillips–Terlevich diagnostic diagram (BPT; J. A. Baldwin et al. 1981; S. Veilleux & D. E. Osterbrock 1987) rather than an AGN-dominated source.

The spectroscopic confirmation of the 61 newly identified GPs is supported by their characteristic rest-frame optical emission-line properties. These objects exhibit the defining spectral signatures of GPs, including strong nebular emission lines, especially the [O III] $\lambda\lambda 4959, 5007$ doublet and prominent Balmer lines, superimposed on relatively faint continua. These features are typical of compact EELGs and provide direct spectroscopic evidence for their GP nature.

To examine the dominant ionization mechanisms of these systems, we placed all 61 newly identified GPs on the BPT diagram, as shown in Figure 3. All 61 spectra have reliable measurements of $\text{H}\beta$, [O III] $\lambda 5007$, $\text{H}\alpha$, and [N II] $\lambda 6583$ and were therefore used in the BPT analysis. The horizontal and vertical axes are defined as $\log([\text{N II}] \lambda 6583/\text{H}\alpha)$ and $\log([\text{O III}] \lambda 5007/\text{H}\beta)$, respectively. Within the uncertainties of the measured emission-line ratios, the vast majority of the confirmed objects lie in the star-forming region of the BPT diagram, below the empirical demarcation line of G. Kauffmann et al. (2003) and the theoretical maximum starburst boundary of L. J. Kewley et al. (2001). Only one source is located slightly above the G. Kauffmann et al. (2003) line but remains below the L. J. Kewley et al. (2001) boundary. This indicates that the optical line emission of the sample is predominantly dominated by photoionization from young massive stars, with no significant evidence for dominant AGN contamination. Therefore, these 61 objects satisfy the high-[O III] emission criterion, while the BPT analysis indicates that the sample as a whole is not dominated by AGN contamination.

Figure 4 presents the rest-frame optical spectra of three representative newly identified GPs. These examples were selected as clear cases to illustrate the defining spectral appearance of the confirmed sample. The spectra show strong nebular emission lines, especially the [O III] $\lambda\lambda 4959, 5007$ doublet and prominent Balmer lines, together with relatively weak continua. These examples provide a direct visual illustration of the spectral characteristics used to confirm the newly identified GP sample.

5.3. Physical Properties and Comparison with Reference Samples

We compared the observational properties of the 61 newly identified GPs with those of the reference GP sample. First, we examined the $\text{EW}([\text{O III}] \lambda 5007)$ distribution, as shown in Figure 5. The newly identified objects show an overall distribution similar to that of the reference sample, with most sources exhibiting large $\text{EW}([\text{O III}] \lambda 5007)$. This behavior is consistent with the strong nebular emission that defines GPs and further supports the reliability of the spectroscopic identifications.

We also compared the r -band magnitude distribution of the 61 newly identified GPs with that of the reference GP sample, as shown in Figure 6. A key finding is that the newly identified GPs extend to fainter magnitudes compared with the reference sample. This implies that some faint compact GPs may have been overlooked by previous selection methods, while our model is capable of recovering such objects from KiDS imaging. Overall, Figure 6 suggests that GP-YOLO11 expands the confirmed GP sample toward a fainter magnitude regime. It not only identifies objects similar to the known GPs but also extends the confirmed sample toward fainter magnitudes, thereby improving the census of this population within the KiDS footprint.

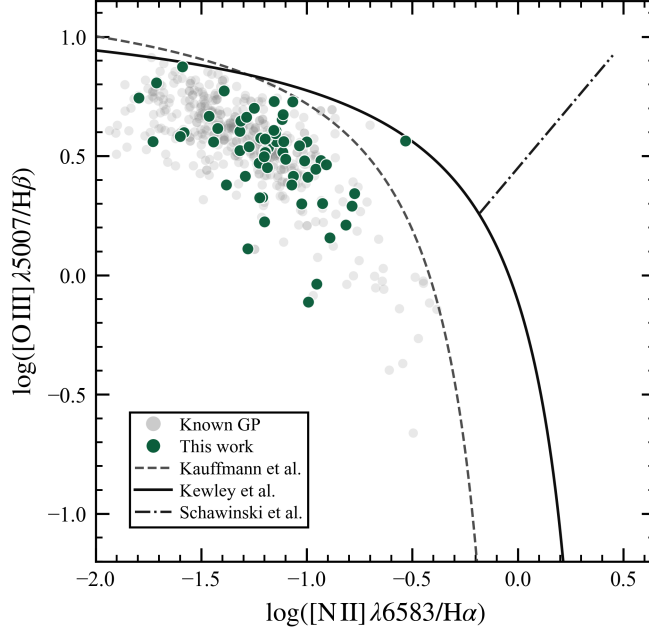


Figure 3. BPT diagram for all 61 newly identified GPs. All plotted objects have reliable measurements of $H\beta$, $[\text{O III}] \lambda 5007$, $H\alpha$, and $[\text{N II}] \lambda 6583$. The horizontal and vertical axes show $\log([\text{N II}] \lambda 6583/H\alpha)$ and $\log([\text{O III}] \lambda 5007/H\beta)$, respectively. The dashed, solid, and dot-dashed curves represent the empirical demarcation of [G. Kauffmann et al. \(2003\)](#), the theoretical maximum starburst boundary of [L. J. Kewley et al. \(2001\)](#), and the AGN-related division from [K. Schawinski et al. \(2007\)](#), respectively.

Finally, Figure 7 compares the redshift distribution of the newly identified GPs with that of the reference GP sample. All confirmed objects lie within the target redshift interval of $0.1 < z < 0.4$, consistent with the selection window adopted in this work. Their distribution overlaps well with that of the reference sample, indicating that the newly identified sources remain within the expected redshift regime of GPs while enlarging the spectroscopically confirmed GP sample in the KiDS footprint.

6. CONCLUSION

We developed and applied GP-YOLO11, a customized deep-learning framework, to conduct a large-scale search for GPs in the KiDS DR5. By analyzing image cutouts constructed from the KiDS multiband data, this approach provides a complementary strategy to traditional photometric color selection and helps reduce the impact of color degeneracies between GPs and compact contaminants.

The proposed framework incorporates two components tailored to the detection of compact astronomical sources. The FAConv module enhances the extraction of local spatial structures, while the NWD loss improves bounding-box regression for small and compact targets. On the independent test set, GP-YOLO11 achieves a Precision of 97.97%, a Recall of 97.61%, and an F1-score of 97.79%, outperforming the traditional color-cut selection, catalog-based machine-learning, image-classification, and object-detection baselines considered in this work.

We applied the trained model to more than one million photometrically pre-selected KiDS DR5 sources within the redshift range of $0.1 < z_B < 0.4$. The final selection yielded a catalog of 5699 high-confidence image-selected GP candidates, which represents the main candidate catalog released in this work. We then cross-matched this catalog with published GP catalogs, SIMBAD, and available archival spectra from DESI and SDSS. This validation recovered 59 previously reported GPs and confirmed 61 objects as newly spectroscopically confirmed GPs that had not previously been reported in the GP literature.

The 61 newly identified GPs show the characteristic strong nebular emission lines of compact EELGs. All of them have reliable measurements of $H\beta$, $[\text{O III}] \lambda 5007$, $H\alpha$, and $[\text{N II}] \lambda 6583$. The vast majority of these objects lie in the star-forming region of the BPT diagram, while one source is located slightly above the empirical line ([G. Kauffmann et al. 2003](#)) but remains below the maximum starburst boundary ([L. J. Kewley et al. 2001](#)). This indicates that the sample is predominantly dominated by photoionization from young massive stars, with no significant evidence for

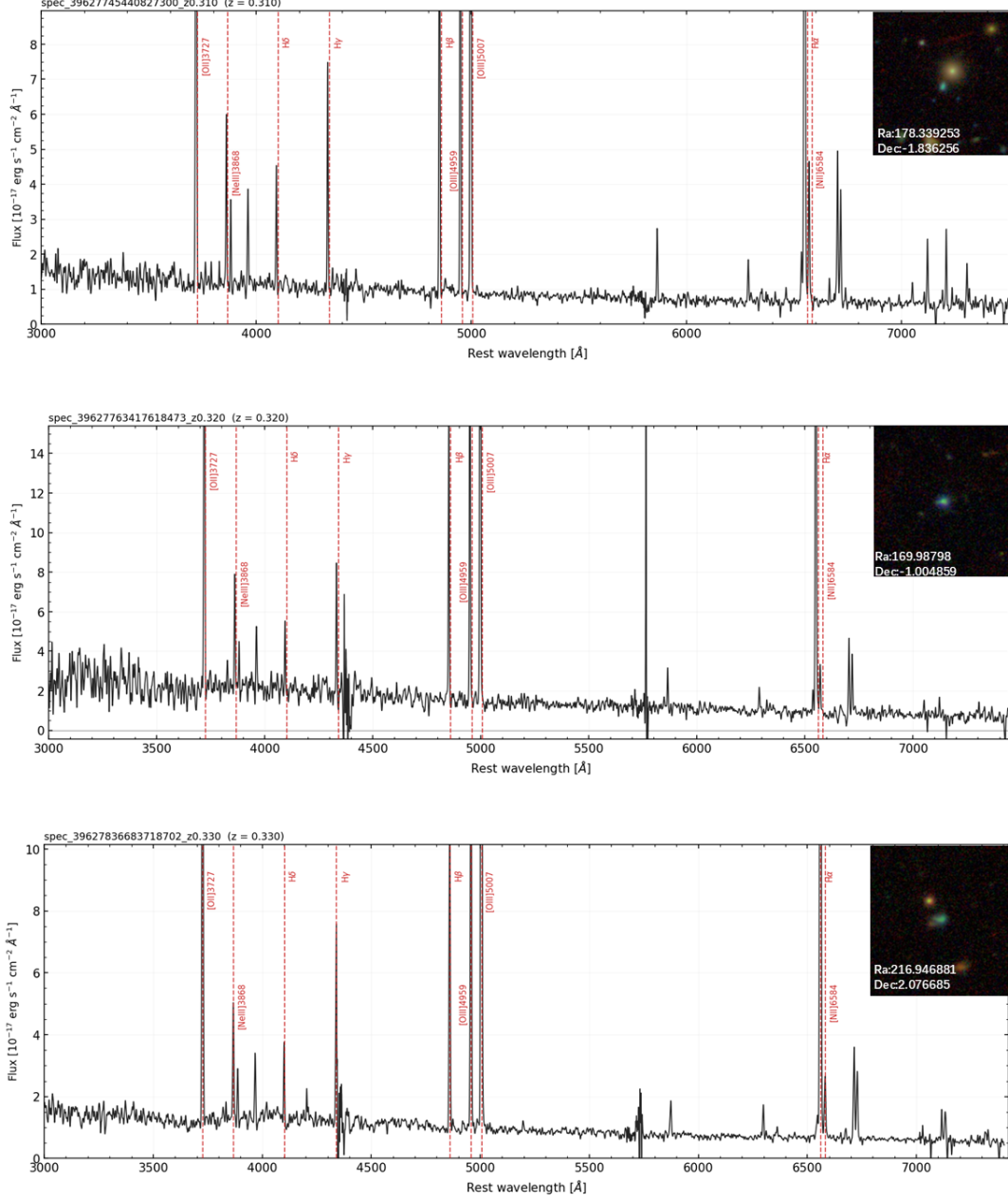


Figure 4. Rest-frame optical spectra of three representative newly identified GPs. The spectra show the characteristic strong nebular emission lines of GP systems, including the [O III] $\lambda\lambda 4959, 5007$ doublet and prominent Balmer lines, superimposed on a relatively faint continuum. These spectral features provide direct confirmation of the extreme star-forming nature of the newly identified objects.

dominant AGN contamination. Their $\text{EW}([\text{O III}] \lambda 5007)$ and redshift distributions are broadly consistent with those of the reference GP sample, while their r -band magnitudes extend to a fainter regime.

These results show that image-based deep learning provides an effective approach for expanding samples of rare compact emission-line galaxies in modern wide-field surveys. The full catalog of 5699 high-confidence image-selected GP candidates provides a useful target list for future spectroscopic follow-up. Because the existing archival spectroscopic coverage from DESI and SDSS does not fully overlap the KiDS footprint, further observations will be important for assessing the nature of the unconfirmed candidates and for characterizing the GP population selected from KiDS imaging.

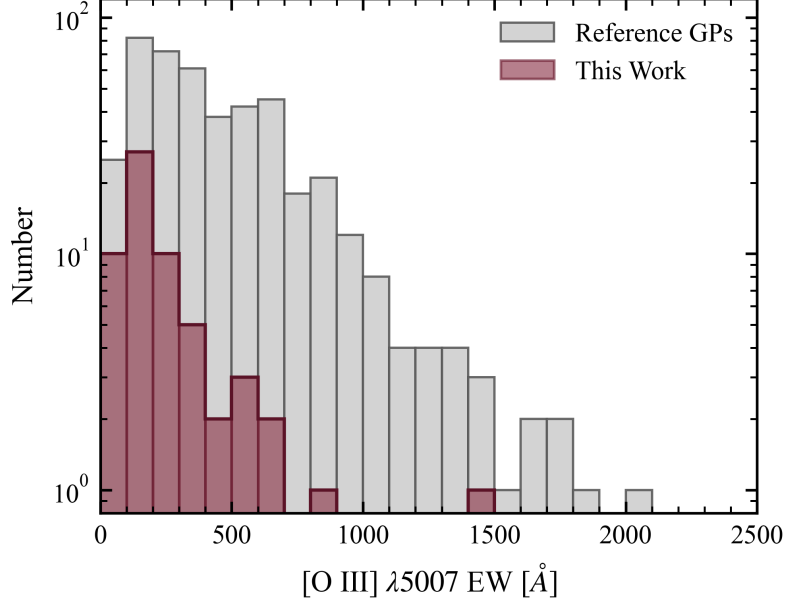


Figure 5. Distribution of [O III] $\lambda 5007$ EW for the newly identified GPs and the reference GP sample. The newly identified objects broadly follow the same overall trend as the reference sample, with most sources showing large [O III] EWs, consistent with the strong nebular emission expected for GP systems.

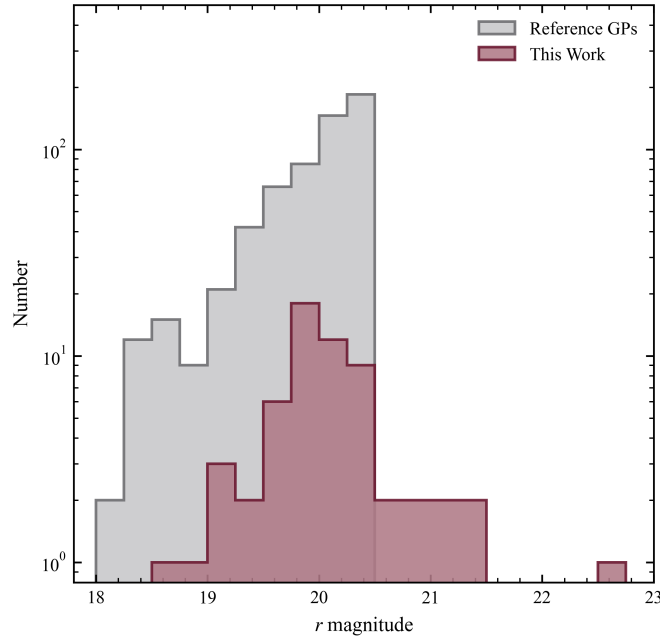


Figure 6. Distribution of KiDS r -band magnitudes for the newly identified GPs and the reference GP sample. The newly identified objects broadly overlap with the reference sample but show a faint tail toward larger r -band magnitudes, indicating that GP-YOLO11 can recover compact GP systems in a fainter photometric regime. The vertical axis is shown on a logarithmic scale.

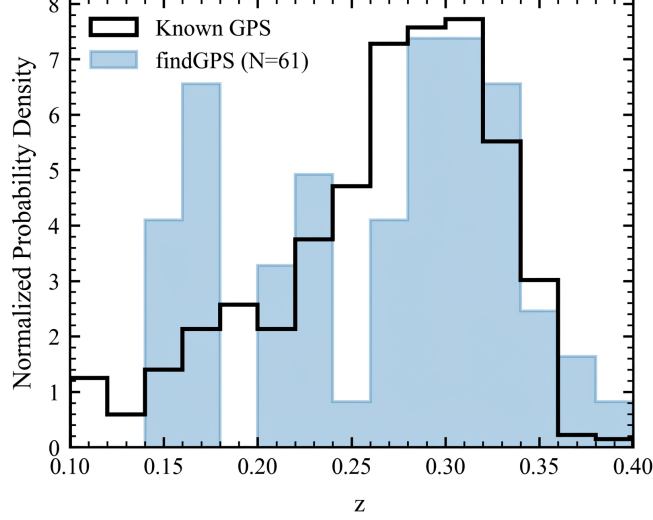


Figure 7. Normalized redshift distribution of the newly identified GPs and the reference GP sample. All confirmed objects lie within the target interval of $0.1 < z < 0.4$, consistent with the selection window adopted in this work. The overlap between the two distributions shows that the newly identified objects remain within the expected redshift regime of GP galaxies.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China under grant Nos. 12273076 and 12133001, the Shenzhen Fundamental Research Program (JCYJ20230807094104009), the National Key R&D Program of China (2025YFA1614101), and the Scientific Research Grants from the China Manned Space Project under grant No. CMS-CSST-2025-A06.

This work is based on imaging data from the Kilo-Degree Survey Data Release 5 (KiDS DR5) and has also made use of archival spectroscopic data from the Dark Energy Spectroscopic Instrument (DESI) and the Sloan Digital Sky Survey (SDSS). We gratefully acknowledge the KiDS, DESI, and SDSS teams for obtaining, processing, and publicly releasing these valuable survey data products.

DATA AVAILABILITY

The imaging data used in this work are publicly available from the KiDS DR5. The archival spectroscopic data used for validation are publicly available from DESI and SDSS databases. The released catalog of 5699 high-confidence GP candidates was constructed by cross-matching the GP-YOLO11-selected coordinates with the KiDS DR5 source catalog, and therefore retains the original KiDS DR5 catalog information together with the model confidence scores and validation labels defined in this work. The catalog of 61 newly identified GPs, the full catalog of 5699 high-confidence GP candidates, the spectra and cutout images of the confirmed sources, and the code used in this paper are available on GitHub: <https://github.com/CXYhaha/GP-YOLO11>.

APPENDIX

A. BASIC PROPERTIES OF THE NEWLY IDENTIFIED GREEN PEA GALAXIES

This appendix presents the basic catalog information for the 61 newly spectroscopically confirmed Green Pea galaxies identified in this work. Table A1 lists their KiDS DR5 source identifiers, coordinates, spectroscopic redshifts, and KiDS multiband photometry. The complete machine-readable catalog, including the full set of matched KiDS DR5 columns, GP-YOLO11 confidence scores, validation labels, spectra, and cutout images, is available through the repository described in the Data Availability section.

Table A1. Basic Properties of the 61 Newly Identified Green Pea Galaxies

No.	KiDS DR5 ID	R.A.	Decl.	z_{spec}	u	g	r	i
		(deg)	(deg)		(mag)	(mag)	(mag)	(mag)
1	KiDSDR5 J135851.502-035311.80	209.714595	-3.886613	0.3700	20.608	20.345	19.844	19.812
2	KiDSDR5 J104031.963-033318.48	160.133180	-3.555135	0.1800	20.536	20.237	19.901	19.901
3	KiDSDR5 J092551.974-015225.14	141.466561	-1.873650	0.2100	20.947	20.715	19.927	20.303
4	KiDSDR5 J120710.117-014725.72	181.792158	-1.790478	0.1800	21.341	20.777	20.337	20.294
5	KiDSDR5 J152623.904-005446.00	231.599603	-0.912779	0.2300	20.356	19.809	19.420	19.229
6	KiDSDR5 J124623.123-003032.57	191.596347	-0.509050	0.3200	20.403	20.063	19.593	20.060
7	KiDSDR5 J115900.286+003710.39	179.751193	0.619554	0.2300	21.361	20.757	20.369	20.299
8	KiDSDR5 J133354.468+003137.68	203.476951	0.527134	0.2700	19.902	19.597	19.156	19.313
9	KiDSDR5 J090454.485+010907.50	136.227021	1.152084	0.3000	22.055	21.100	20.128	20.102
10	KiDSDR5 J103932.320+011614.48	159.884667	1.270689	0.2700	21.234	20.616	20.094	20.100
11	KiDSDR5 J113533.655+021443.92	173.890230	2.245534	0.1700	20.601	20.338	19.940	19.904
12	KiDSDR5 J142530.454+030644.29	216.376894	3.112305	0.1800	19.967	19.606	19.270	19.285
13	KiDSDR5 J134511.961+033208.25	206.299841	3.535627	0.2600	21.023	20.495	20.117	20.023
14	KiDSDR5 J130626.386-013535.14	196.609943	-1.593096	0.1400	21.025	20.391	20.172	20.007
15	KiDSDR5 J125646.614-032841.63	194.194228	-3.478232	0.1400	20.510	19.791	19.499	19.300
16	KiDSDR5 J112231.535-031345.26	170.631398	-3.229239	0.3600	20.846	20.512	19.959	20.017
17	KiDSDR5 J111456.419-022757.26	168.735080	-2.465908	0.1400	21.204	20.572	20.297	20.053
18	KiDSDR5 J115321.420-015010.52	178.339253	-1.836256	0.3100	21.478	21.095	20.522	20.875
19	KiDSDR5 J141302.381-013352.91	213.259922	-1.564699	0.3700	21.360	21.079	20.494	20.444
20	KiDSDR5 J114936.173-010938.23	177.400723	-1.160621	0.3300	21.273	21.091	20.514	21.245
21	KiDSDR5 J111957.115-010017.49	169.987980	-1.004859	0.3200	20.716	20.477	20.099	20.369
22	KiDSDR5 J144302.226-004402.52	220.759277	-0.734034	0.3100	21.151	20.352	19.284	19.478
23	KiDSDR5 J120331.183-000019.99	180.879932	-0.005553	0.1800	21.392	20.740	20.355	20.332
24	KiDSDR5 J143240.938+003917.76	218.170577	0.654935	0.1800	20.609	20.448	19.875	20.091
25	KiDSDR5 J121253.933+014031.76	183.224721	1.675490	0.3800	21.107	20.844	20.396	20.323
26	KiDSDR5 J125731.416+015600.14	194.380900	1.933373	0.2400	21.344	20.737	20.086	20.171
27	KiDSDR5 J114048.760+030259.34	175.203169	3.049818	0.2800	20.781	20.417	20.071	20.210
28	KiDSDR5 J104817.384+030938.41	162.072435	3.160671	0.2800	20.544	20.219	19.932	19.984
29	KiDSDR5 J131847.097+033656.66	199.696238	3.615741	0.1800	20.261	19.746	19.304	19.198
30	KiDSDR5 J131132.023+034743.76	197.883430	3.795489	0.2900	21.176	20.613	20.094	20.076
31	KiDSDR5 J134046.796-030240.05	205.194987	-3.044461	0.3600	21.344	21.094	20.425	20.639
32	KiDSDR5 J144808.674-002138.21	222.036143	-0.360615	0.3000	21.532	20.943	20.341	20.585
33	KiDSDR5 J144459.853-002106.89	221.249390	-0.351916	0.3300	22.666	22.382	21.354	23.046
34	KiDSDR5 J142747.251+020436.06	216.946881	2.076685	0.3300	21.527	21.035	20.381	20.597
35	KiDSDR5 J141940.051+032010.48	214.916880	3.336246	0.1800	20.594	20.307	19.844	19.871
36	KiDSDR5 J142741.972+031208.07	216.924885	3.202243	0.3100	20.970	20.591	20.248	20.447
37	KiDSDR5 J140530.107+035208.46	211.375446	3.869019	0.2800	21.403	20.922	20.438	20.452
38	KiDSDR5 J123204.546-024339.56	188.018945	-2.727658	0.2700	20.167	19.821	19.469	19.687
39	KiDSDR5 J132942.860+024724.11	202.428587	2.790032	0.2200	21.911	21.354	20.850	20.916

Table A1 continued on next page

Table A1 (continued)

No.	KiDS DR5 ID	R.A.	Decl.	z_{spec}	u	g	r	i
		(deg)	(deg)		(mag)	(mag)	(mag)	(mag)
40	KiDSDR5 J133142.116+035209.71	202.925484	3.869366	0.1600	22.207	21.646	21.272	21.275
41	KiDSDR5 J110915.362-012851.99	167.314009	-1.481109	0.3600	21.011	20.662	20.115	20.216
42	KiDSDR5 J113203.381+020556.45	173.014090	2.099016	0.2600	21.396	20.858	20.426	20.442
43	KiDSDR5 J112209.089+021201.13	170.537873	2.200314	0.2400	21.500	21.140	20.659	20.736
44	KiDSDR5 J090543.532+001950.05	136.431386	0.330572	0.3200	21.168	20.869	20.401	20.612
45	KiDSDR5 J092152.505+010443.89	140.468772	1.078861	0.3100	22.297	21.892	21.416	21.517
46	KiDSDR5 J115502.517+012316.88	178.760489	1.388023	0.2200	22.429	21.937	21.399	21.467
47	KiDSDR5 J153811.088-013310.88	234.546200	-1.553024	0.3200	21.273	20.919	20.423	20.464
48	KiDSDR5 J151235.812+002100.14	228.149219	0.350041	0.3300	22.380	22.002	21.165	21.680
49	KiDSDR5 J152503.953+013704.81	231.266472	1.618005	0.3000	21.413	20.847	20.281	20.380
50	KiDSDR5 J152855.280+024042.83	232.230336	2.678564	0.3100	20.267	19.987	19.591	19.649
51	KiDSDR5 J155106.235+020508.09	237.775982	2.085583	0.3300	22.112	21.604	20.892	21.097
52	KiDSDR5 J122224.608-030924.27	185.602535	-3.156743	0.2500	19.514	19.117	18.783	18.702
53	KiDSDR5 J124548.674-021722.17	191.452809	-2.289494	0.3400	21.236	20.940	20.434	20.537
54	KiDSDR5 J150130.656-014633.95	225.377735	-1.776099	0.2300	20.562	20.096	19.725	19.636
55	KiDSDR5 J112828.249-004152.39	172.117708	-0.697887	0.2400	22.472	21.865	21.381	21.400
56	KiDSDR5 J131238.683-000150.54	198.161180	-0.030707	0.2900	20.910	20.508	20.022	20.151
57	KiDSDR5 J115515.874+001824.53	178.816145	0.306815	0.2100	21.700	21.585	20.718	21.298
58	KiDSDR5 J112958.087+002928.01	172.492033	0.491114	0.3300	21.296	20.712	20.156	20.159
59	KiDSDR5 J111306.555+020427.00	168.277315	2.074167	0.3300	21.935	21.553	20.900	21.466
60	KiDSDR5 J113700.424+021243.65	174.251767	2.212126	0.1400	21.128	20.483	20.207	20.000
61	KiDSDR5 J131129.505+021344.81	197.872941	2.229115	0.3000	20.699	20.338	19.802	20.163

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